



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Utkarsh Raj
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodologies

- Data Collection via Web scraping
- Data Wrangling
- Complete the EDA with Visualization
- Dashboard

All results

- Predict first stage of the Falcon 9 lands successfully

Introduction

Project background and context

- Space Y wants to launch rockets with minimal costs

Problems you want to find answers

- Determine the price of each launch
- Determine if SpaceX will reuse the first stage

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Using SpaceX API and Web scraping
- Perform data wrangling
 - Use flowcharts and key phrases
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Built, evaluated, improved, and found the best classification model

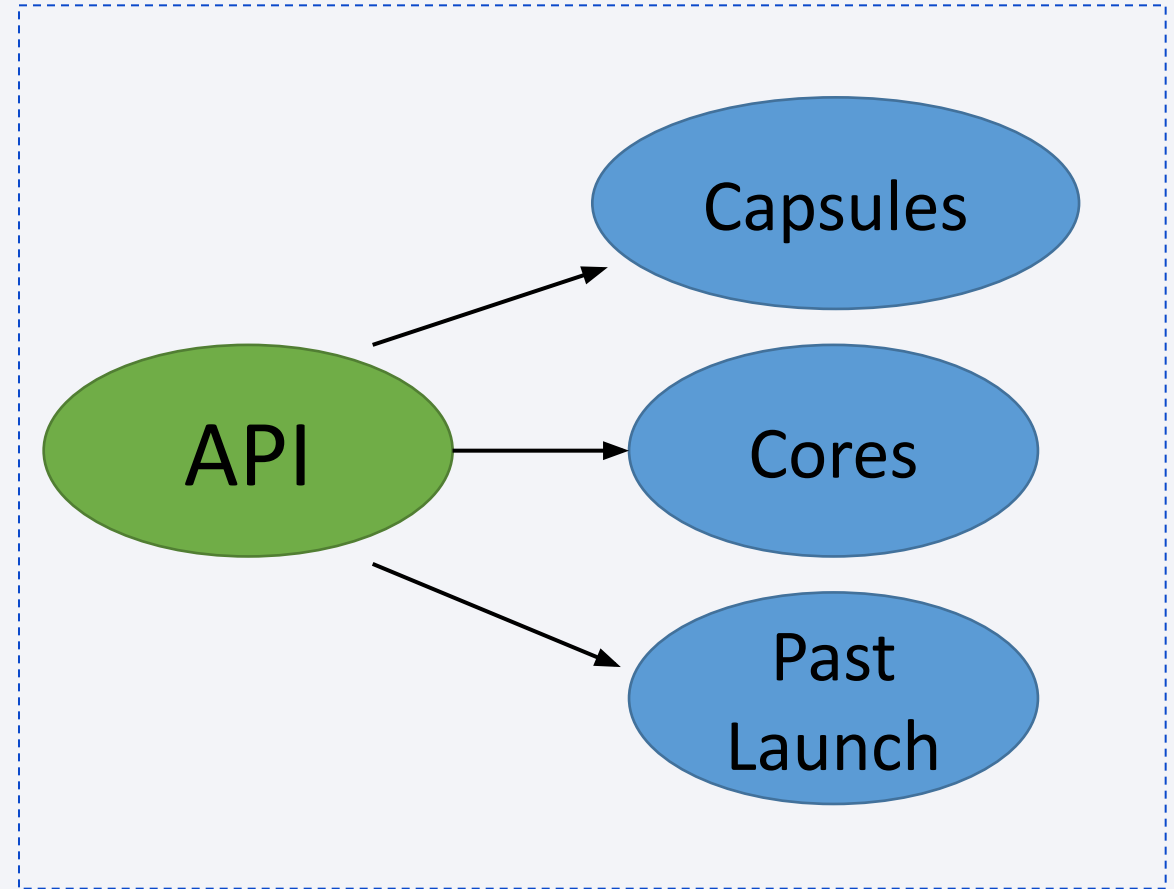
Data Collection

Describe how data sets were collected

- API will give us data about launches, including information about the rocket used, payload delivered, launch specifications, landing specifications, and landing outcome
- Web scrape some HTML tables that contain valuable Falcon 9 launch records

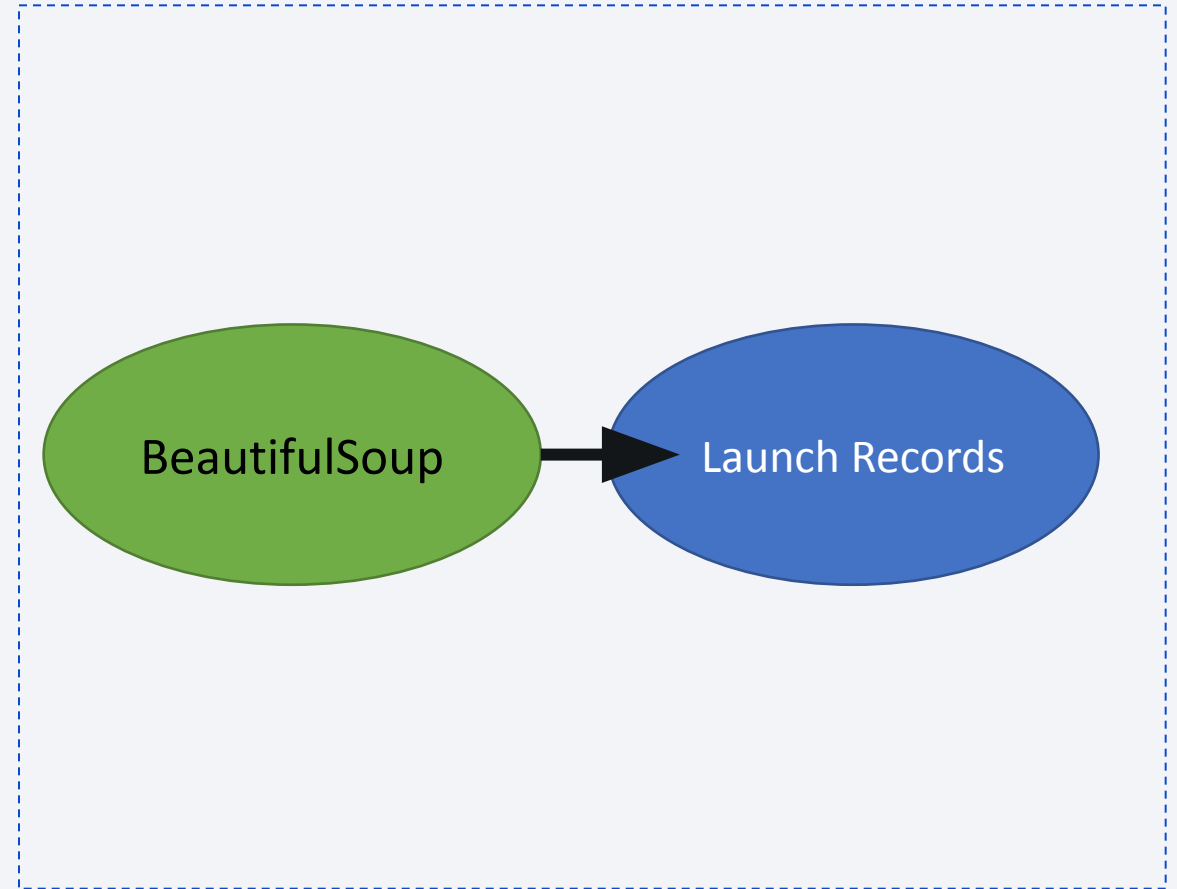
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook ([must include completed code cell and outcome cell](#)), as an external reference and peer-review purpose



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose



Data Wrangling

Describe how data were processed

- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurrence of mission outcome per orbit type
- Create a landing outcome label from outcome column
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

EDA with Data Visualization

Summarize what charts were plotted and why you used those charts

- Visualize the relationship between Flight Number and Launch Site
- Visualize the relationship between Payload and Launch Site
- Visualize the relationship between success rate of each orbit type
- Visualize the relationship between FlightNumber and Orbit type
- Visualize the relationship between Payload and Orbit type
- Visualize the launch success yearly trend
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

EDA with SQL

Summarize the SQL queries you performed

- Retrieve the most recent date from the SpaceX table
- Display the minimum payload mass
- Total payload_mass_kg carried by the booster versions
- Display 5 records launched on Friday
- Unique launch sites

Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

Build an Interactive Map with Folium

Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map

- Mark all launch sites on a map
- Mark the success/failed launches for each site
- Calculate the distances between a launch site to its proximities

Explain why you added those objects

Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

Summarize what plots/graphs and interactions you have added to a dashboard

- Analyzing launch site geo and proximities
- Choose an optimal launch site

Explain why you added those plots and interactions

Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

Summarize how you built, evaluated, improved, and found the best performing classification model

Create 2 classes

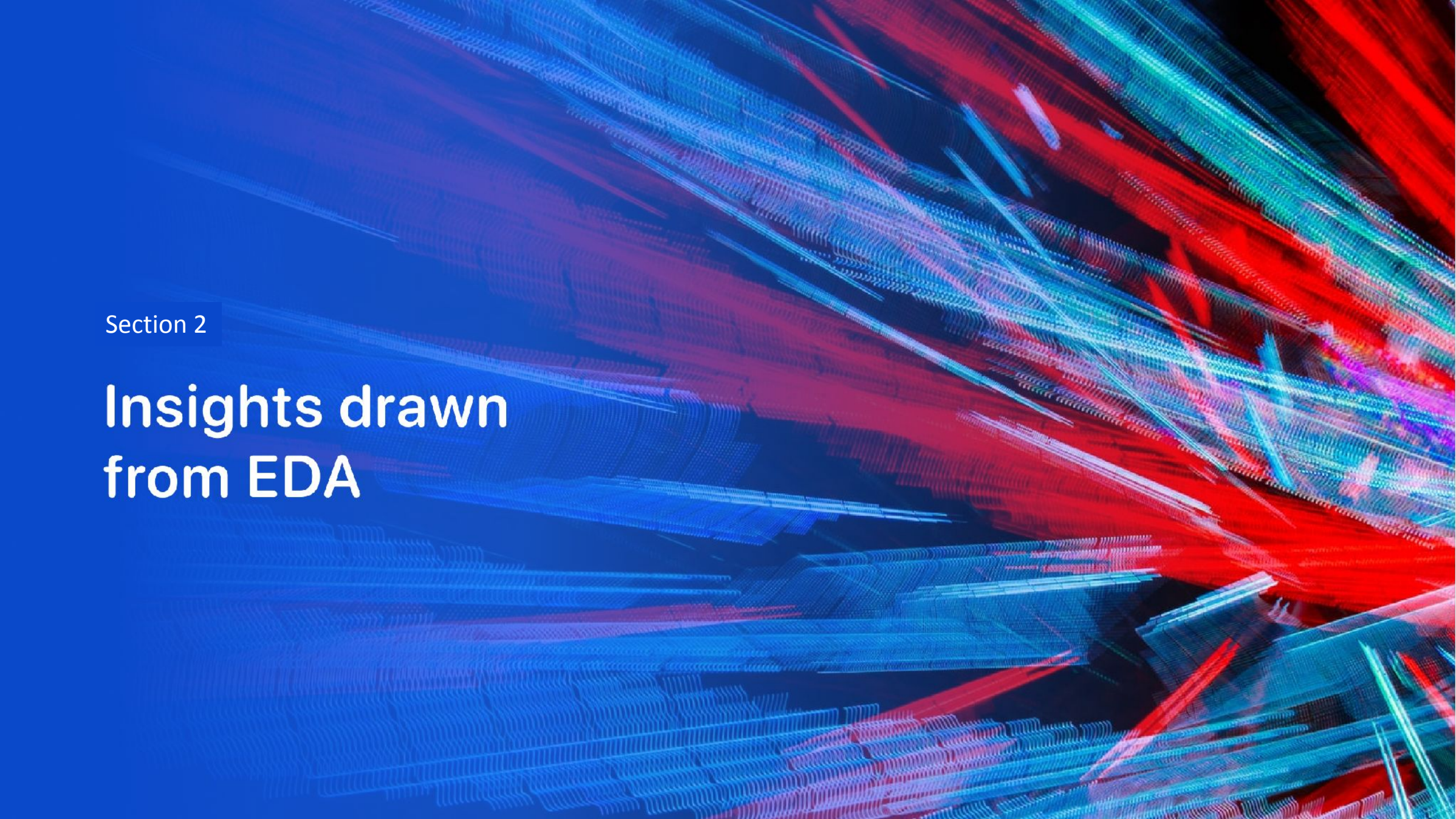
Standardize the data, create a logistic regression object then create a GridSearchCV.

You need present your model development process using key phrases and flowchart

Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

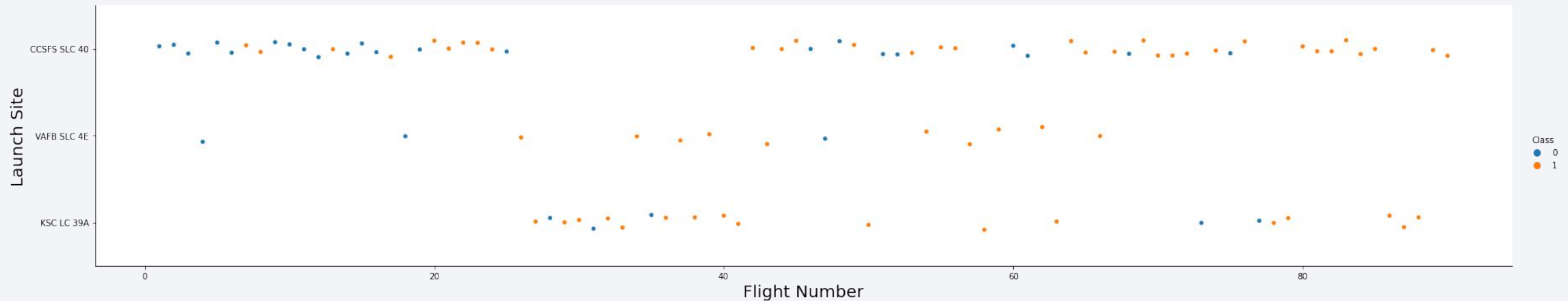
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and digital complexity.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

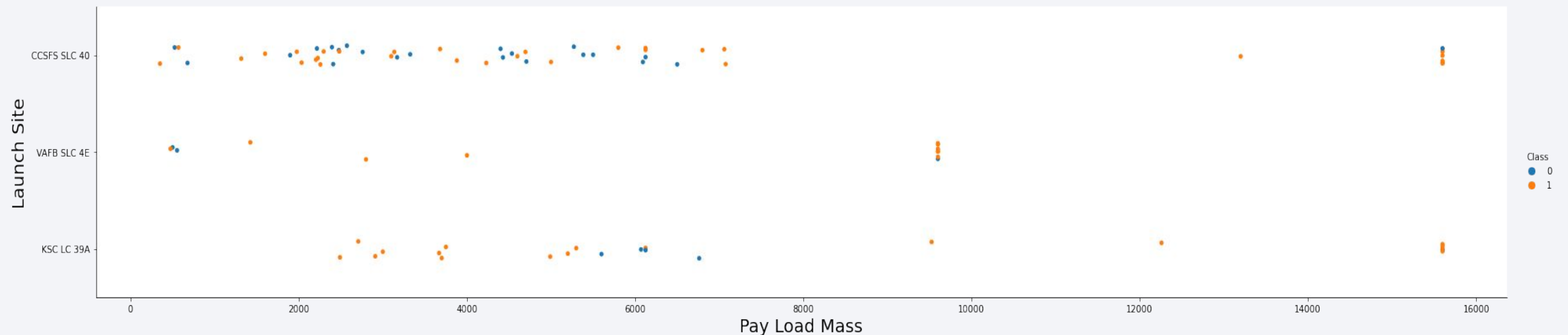
- Show a scatter plot of Flight Number vs. Launch Site



Show the screenshot of the scatter plot with explanations

Payload vs. Launch Site

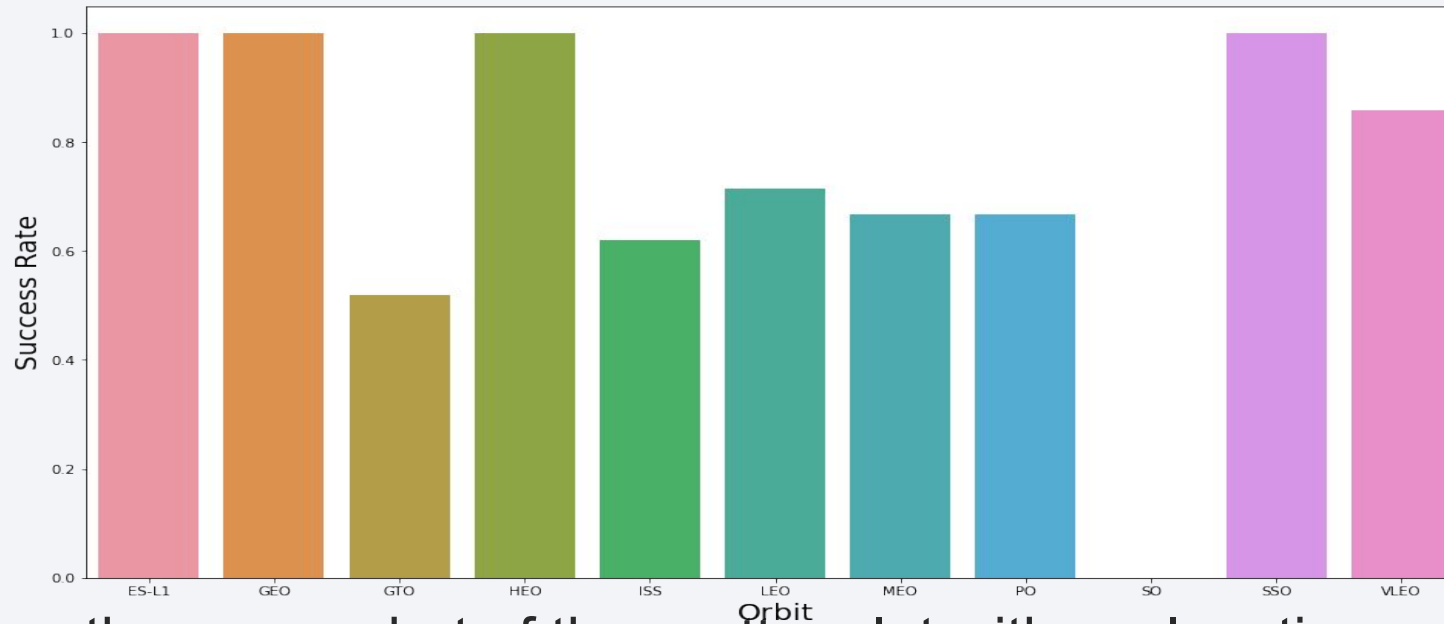
- Show a scatter plot of Payload vs. Launch Site



- Show the screenshot of the scatter plot with explanations

Success Rate vs. Orbit Type

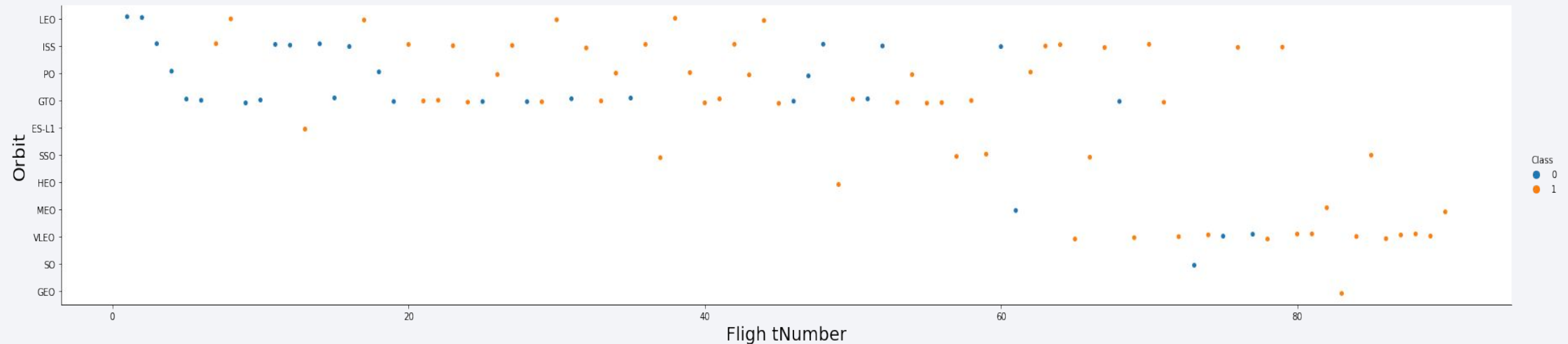
- Show a bar chart for the success rate of each orbit type



Show the screenshot of the scatter plot with explanations

Flight Number vs. Orbit Type

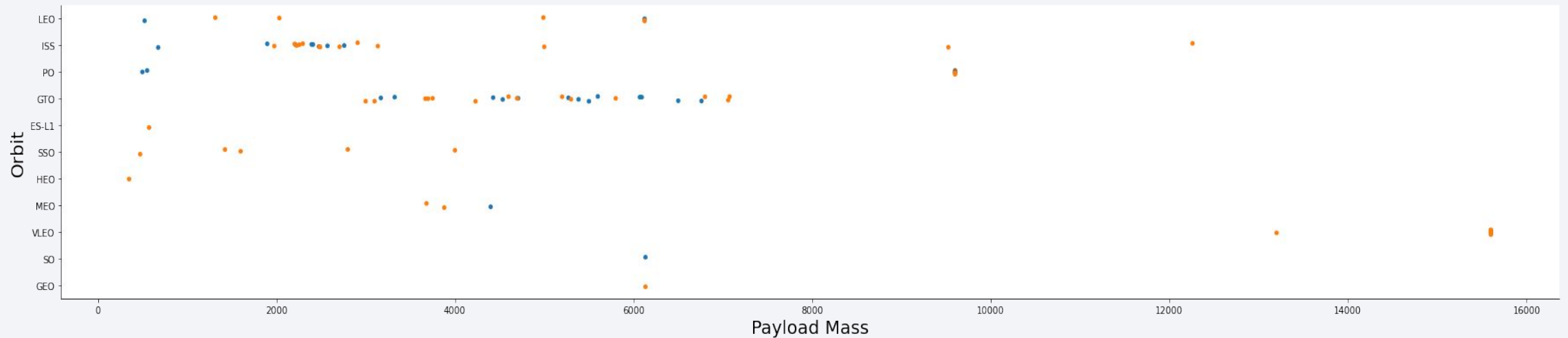
- Show a scatter point of Flight number vs. Orbit type



- Show the screenshot of the scatter plot with explanations

Payload vs. Orbit Type

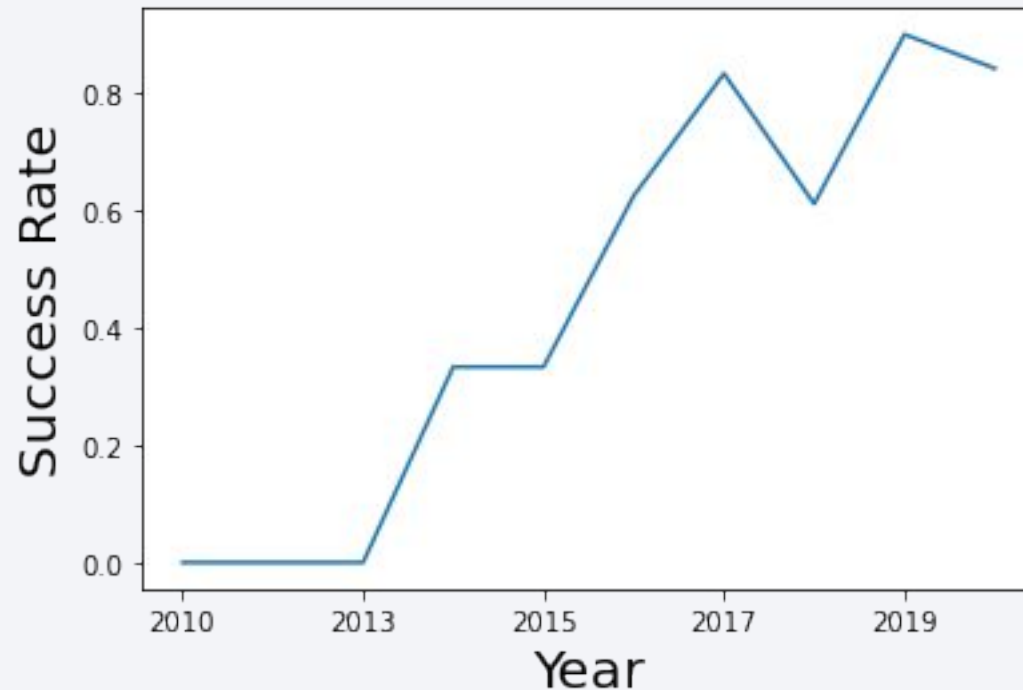
- Show a scatter point of payload vs. orbit type



- Show the screenshot of the scatter plot with explanations

Launch Success Yearly Trend

- Show a line chart of yearly average success rate



- Show the screenshot of the scatter plot with explanations

All Launch Site Names

- Find the names of the unique launch sites

Display the names of the unique launch sites in the space mission

```
In [16]: pd.read_sql_query("SELECT DISTINCT Launch_Site FROM SPACEX \
                          LIMIT 5;", db)
```

Out[16]:

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

- Present your query result with a short explanation here

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
In [17]: pd.read_sql_query("SELECT * FROM SPACEX \nWHERE Launch_Site LIKE 'CCA%' \nLIMIT 5;", db)
```

Out[17]:

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	04-06-2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	08-12-2010	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	22-05-2012	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	08-10-2012	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	01-03-2013	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Present your query result with a short explanation here

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [18]: pd.read_sql_query("SELECT SUM (PAYLOAD_MASS_KG_) FROM SPACEX \n\n                WHERE Customer LIKE '%NASA (CRS)%';", db)
```

```
Out[18]:
```

SUM (PAYLOAD_MASS_KG_)	
0	48213

- Present your query result with a short explanation here

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
In [19]: pd.read_sql_query("SELECT AVG (PAYLOAD_MASS_KG_) FROM SPACEX \n\n                WHERE Booster_Version = 'F9 v1.1';", db)
```

```
Out[19]:
```

	AVG (PAYLOAD_MASS_KG_)
0	2928.4

- Present your query result with a short explanation here

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
In [21]: df[df["Landing_Outcome"] == "Success (ground pad)"].head(3)
```

```
Out[21]:
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
19	22-12-2015	01:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)
26	18-07-2016	04:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
29	19-02-2017	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)

- Present your query result with a short explanation here

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
In [23]: df2[(df2["PAYLOAD_MASS_KG_"] > 4000) & (df2["PAYLOAD_MASS_KG_"] <= 6000)]
```

Out[23]:	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
23	06-05-2016	05:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
27	14-08-2016	05:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
31	30-03-2017	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
42	11-10-2017	22:53:00	F9 FT B1031.2	KSC LC-39A	SES-11 / EchoStar 105	5200	GTO	SES EchoStar	Success	Success (drone ship)

- Present your query result with a short explanation here

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

List the total number of successful and failure mission outcomes

```
In [24]: df["Mission_Outcome"].value_counts()
```

```
Out[24]: Success                98  
Failure (in flight)             1  
Success (payload status unclear) 1  
Success                        1  
Name: Mission_Outcome, dtype: int64
```

- Present your query result with a short explanation here

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload

m

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
In [25]: df["PAYLOAD_MASS_KG_"].max()
```

```
Out[25]: 15600
```

```
In [26]: df[df["PAYLOAD_MASS_KG_"] == 15600]
```

```
Out[26]:
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
74	11-11-2019	14:56:00	F9 B5 B1048.4	CCAFS SLC-40	Starlink 1 v1.0, SpaceX CRS-19	15600	LEO	SpaceX	Success	Success
77	07-01-2020	02:33:00	F9 B5 B1049.4	CCAFS SLC-40	Starlink 2 v1.0, Crew Dragon in-flight abort t...	15600	LEO	SpaceX	Success	Success
79	29-01-2020	14:07:00	F9 B5 B1051.3	CCAFS SLC-40	Starlink 3 v1.0, Starlink 4 v1.0	15600	LEO	SpaceX	Success	Success
80	17-02-2020	15:05:00	F9 B5 B1056.4	CCAFS SLC-40	Starlink 4 v1.0, SpaceX CRS-20	15600	LEO	SpaceX	Success	Failure
82	18-03-2020	12:16:00	F9 B5 B1048.5	KSC LC-39A	Starlink 5 v1.0, Starlink 6 v1.0	15600	LEO	SpaceX	Success	Failure
83	22-04-2020	19:30:00	F9 B5 B1051.4	KSC LC-39A	Starlink 6 v1.0, Crew Dragon Demo-2	15600	LEO	SpaceX	Success	Success
85	04-06-2020	01:25:00	F9 B5 B1049.5	CCAFS SLC-40	Starlink 7 v1.0, Starlink 8	15600	LEO	SpaceX, Planet Labs	Success	Success

- Present your query result with a short explanation here

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
In [32]: df3 = df[(df["Landing_Outcome"] == "Failure (drone ship)") & (df["Year"] == 2015)]
df3
```

```
Out[32]:
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome	Year
13	2015-10-01	09:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)	2015
16	2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)	2015

- Present your query result with a short explanation here

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
In [33]: df4 = df[df["Date"].between("2010-04-06", "2017-03-20")]
```

```
In [34]: df4["Landing_Outcome"].value_counts()
```

```
Out[34]: No attempt          10  
Failure (drone ship)        5  
Success (ground pad)        5  
Success (drone ship)        5  
Controlled (ocean)          3  
Failure (parachute)          2  
Uncontrolled (ocean)         2  
Precluded (drone ship)       1  
Name: Landing_Outcome, dtype: int64
```

- Present your query result with a short explanation here

A satellite view of Earth from space, showing the curvature of the planet and the glow of city lights at night. The lights are concentrated in the lower right portion of the frame, with a dark blue atmosphere above. The background is a deep black space with some distant stars visible.

Section 3

Launch Sites Proximities Analysis

Mark all launch sites on a map

- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map

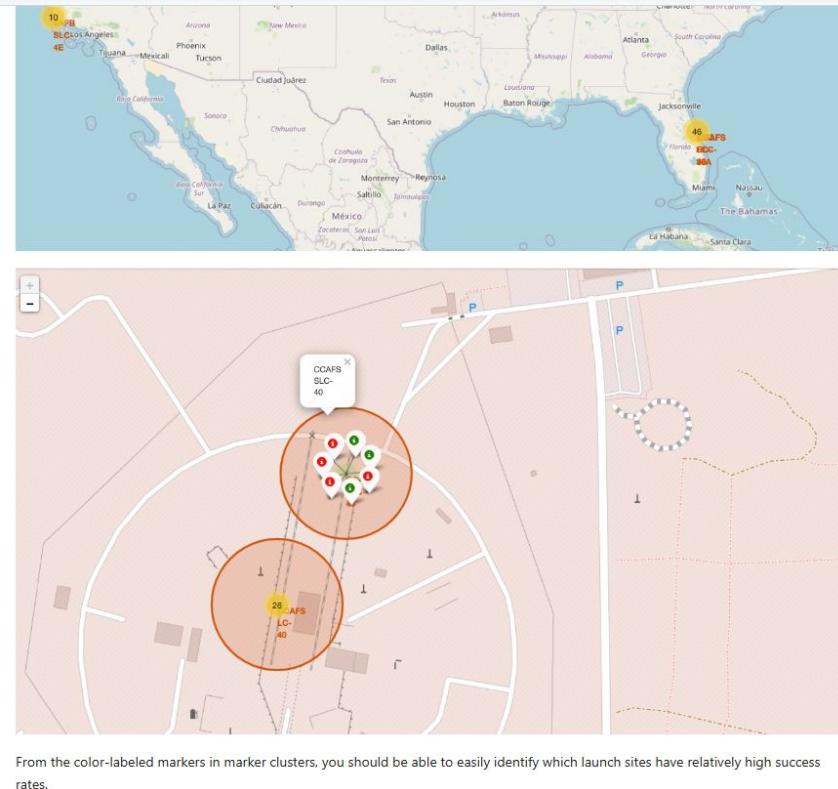
The generated map with marked launch sites should look similar to the following:



- Explain the important elements and findings on the screenshot

Mark the success/failed launches for each site on the map

- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map



- Explain the important elements and findings on the screenshot

Calculate the distances between a launch site to its proximities

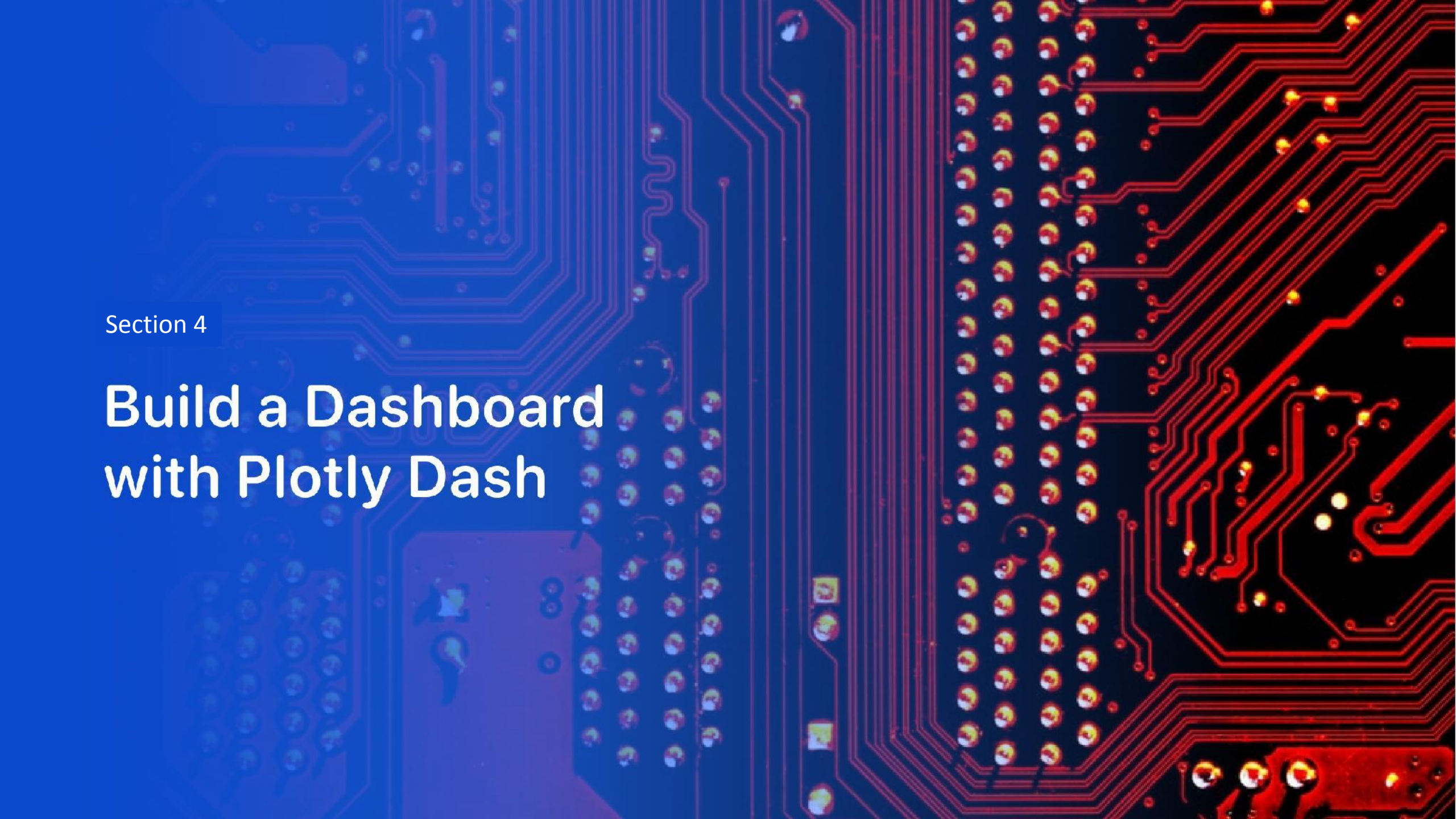
- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed

Your updated map with distance line should look like the following screenshot:



TODO: Similarly, you can draw a line between a launch site to its closest city, railway, highway, etc. You need to use `MousePosition` to find the their coordinates on the map first

- Explain the important elements and findings on the screenshot

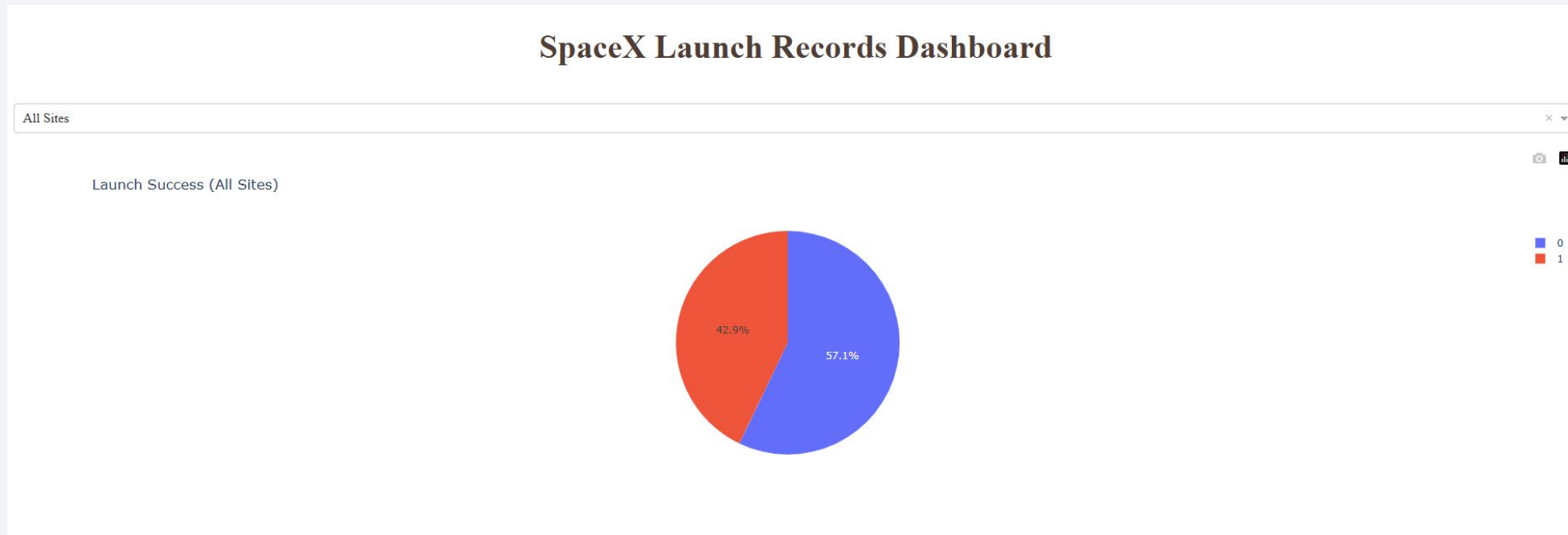


Section 4

Build a Dashboard with Plotly Dash

Launch success count for all sites

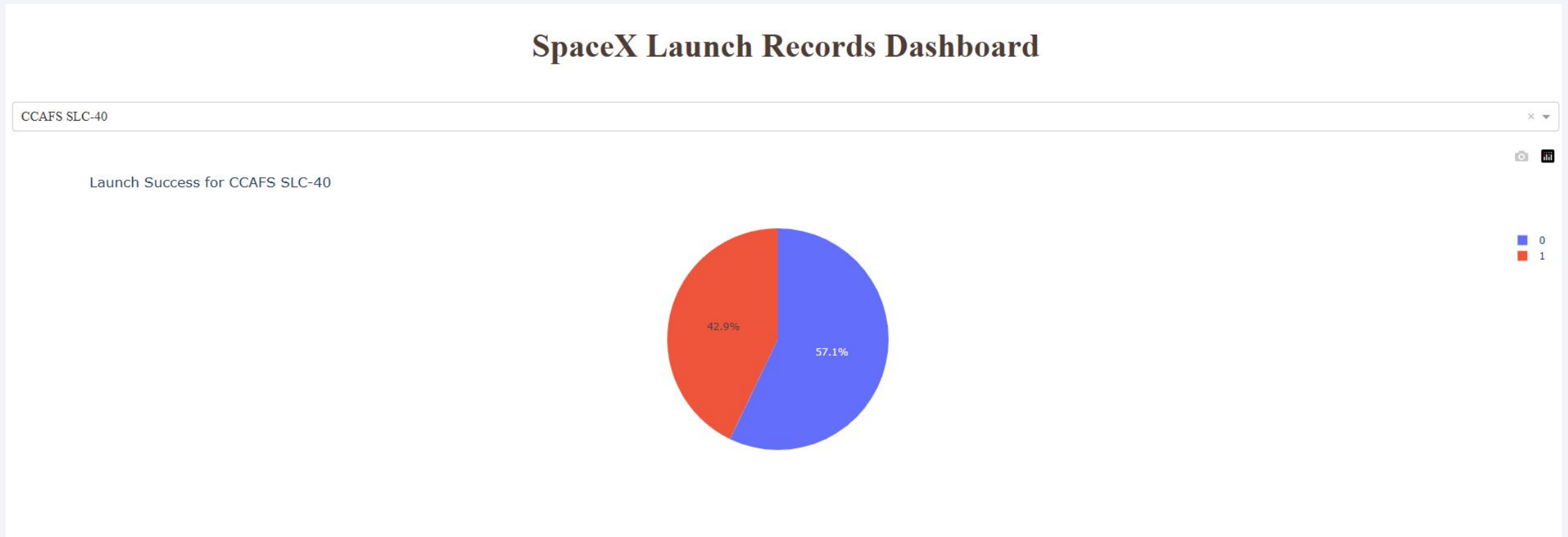
- Show the screenshot of launch success count for all sites, in a piechart



- Explain the important elements and findings on the screenshot

Launch site with highest launch success ratio

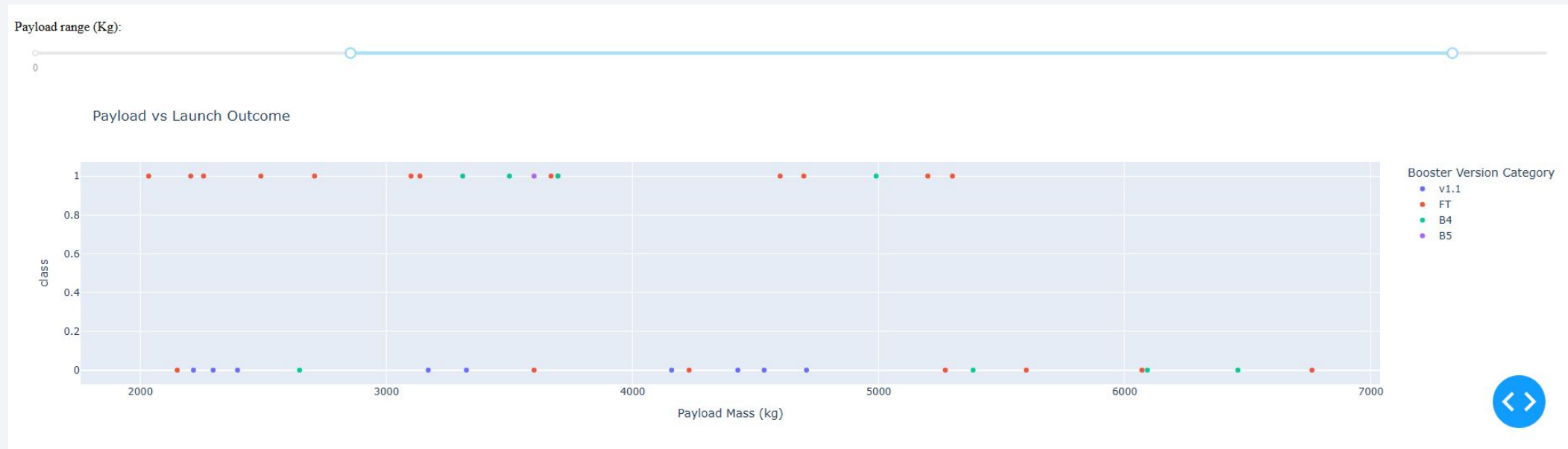
- Show the screenshot of the piechart for the launch site with highest launch success ratio



- Explain the important elements and findings on the screenshot

Payload vs. Launch Outcome scatter plot for all sites

- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider



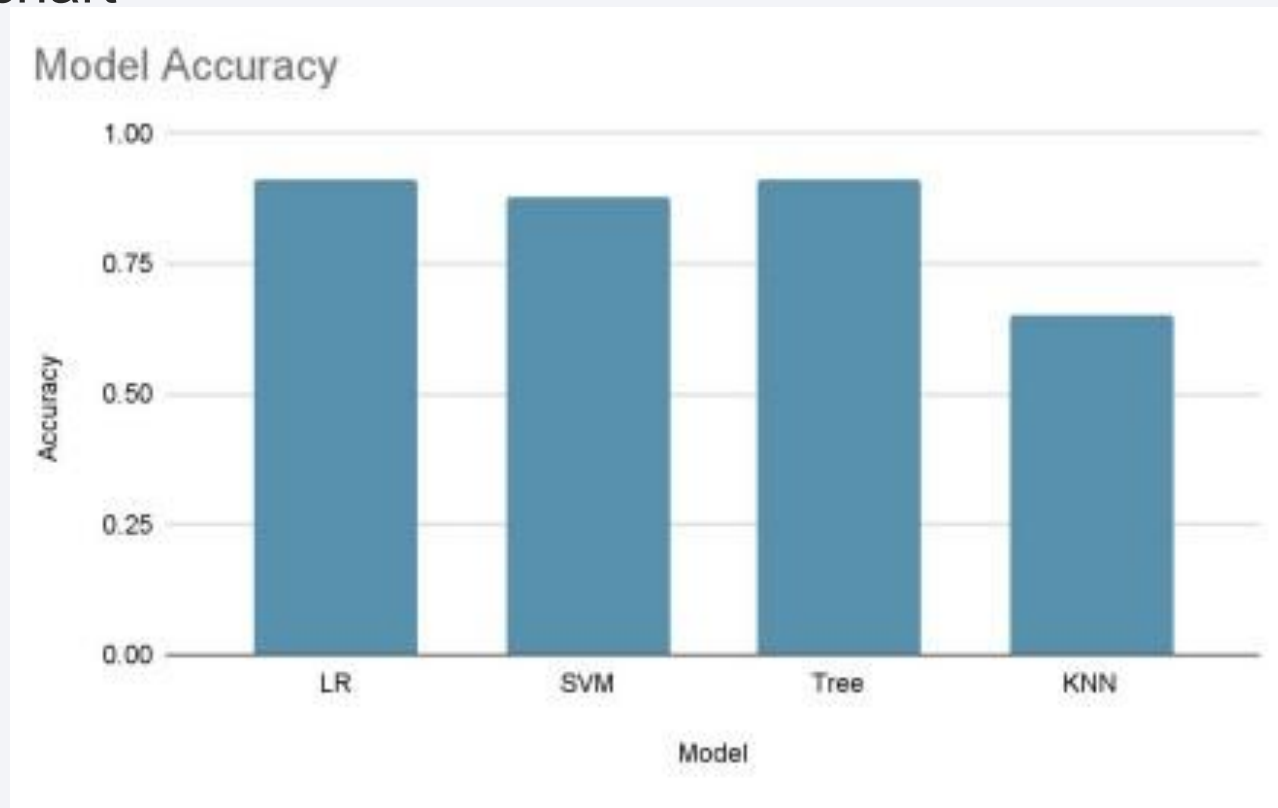
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart



- Find which model has the highest classification accuracy

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation

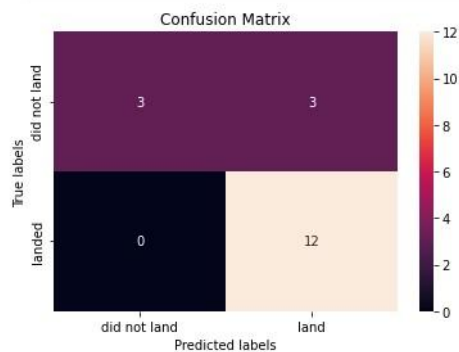
Calculate the accuracy on the test data using the method `score` :

```
In [29]: logreg_cv.score(X=X_test, y=y_test)
```

```
Out[29]: 0.8333333333333334
```

Lets look at the confusion matrix:

```
In [30]: yhat=logreg_cv.predict(X_test)
plot_confusion_matrix(y_test,yhat)
```



Examining the confusion matrix, we see that logistic regression can distinguish between the different classes. We see that the major problem is false positives.

Conclusions

- Predicting Falcon 9's first-stage landing success can optimize SpaceX's operations and reduce launch costs.
- By leveraging historical data, we can develop accurate models to forecast landing outcomes.
- This prediction model provides valuable insights into reusability, benefiting both SpaceX and its competitors.
- With continuous improvement, this model could revolutionize cost estimation and efficiency in spaceflight.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

